

TIME SERIES ANALYSIS AND ARIMA FORECASTING OF MALAYSIA WEEKLY RETAIL FUEL PRICES (RON95)

PREDICTIVE ANALYSIS MDA 3083

SEMESTER II 2025/2026

Prepared by:

Kauthar Binti Mohd Hisham (S76156)

Nur Ayuni Binti Azizi (S76255)

Nurul Dyaasykin Binti Mohd Ezany (S76282)

Nurul Anis Binti Ahmad Saiful (S76279)

Danish Hateem Binti Jumat (S74742)

Lecturer:

Sharifah Sakinah

1.0 INTRODUCTION

Fuel prices are among the most significant economic indicators in Malaysia, as they have a direct impact on transportation costs, household expenditures, and the prices of goods and services throughout the economy. Malaysia's retail fuel pricing covers several grades and types, including RON95, as well as diesel, which is priced differently for Peninsular Malaysia and East Malaysia (Sabah, Sarawak, and Labuan). These fuel prices were heavily regulated through government subsidies and price-control mechanisms. However, the implementation of a managed float and targeted subsidy framework in mid-2025 introduced weekly fuel price revisions that are more responsive to fluctuations in global crude oil prices and exchange rate movements. Accurate analysis and forecasting of retail fuel prices are essential for policymakers, businesses, and consumers to support effective planning and decision-making. Therefore, this study examines the historical pattern of a variable over time, identify trend and seasonal components, and build forecasting models such as the Autoregressive Integrated Moving Average (ARIMA) model.

2.0 OBJECTIVES

The objectives of this study are :

1. To analyze the historical trends and patterns of RON95 fuel prices in Malaysia using time series data.
2. To identify an appropriate ARIMA model for forecasting future RON95 fuel prices based on historical observations.
3. To examine the accuracy and effectiveness of the ARIMA forecasting model in predicting future RON95 fuel prices.

3.0 DATA DESCRIPTION

3.1 DATA SOURCE

The dataset used in this study is the "Weekly Retail Fuel Price dataset, obtained from the official Malaysia Open Data portal, data.gov.my, which is published by the Ministry of Finance.

3.2 DATA DESCRIPTION

Variable	Description
date	The date (week ending) on which the fuel price was effective, in YYYY-MM-DD format.
ron95	Retail price of RON95 petrol in Malaysian Ringgit (RM) per litre - the main variable of interest for this study.
ron97	Retail price of RON97 petrol (RM/litre).
diesel	Retail price of diesel for Peninsular Malaysia (RM/litre).
diesel_eastmsia	Retail price of diesel for East Malaysia (Sabah, Sarawak, Labuan) (RM/litre).

3.3 VARIABLES OF INTEREST

The primary variable analysed in this study is ron95 (RON95 retail price in RM/litre), recorded weekly. This satisfies the minimum dataset requirement of having a clearly defined time variable (date) and more than 100 observations (464 weekly observations after cleaning).

3.4 SOFTWARE

The software tools were used in this study is Phyton (Jupyter Notebook) and Excel, in line with the assignment requirement :

- Phyton (Jupyter Notebook) – used for data exploration, data cleaning, ARIMA model and forecasting
- Power BI – we use Power BI for the dashboard

4.0 METHODOLOGY

4.1 DATA PREPROCESSING (PHYTON)

The raw dataset was imported and explored in Phyton using Jupyter Notebook :

- The dataset was filtered to retain only records where series_type = "level", representing actual price values rather than weekly changes.
- The date column was changed into a proper datetime format and set as the index of the dataset to support time series analysis.
- Unnecessary variables, including series_type, diesel_eastmsia, ron95_budi95, and ron95_skps, were removed since they were not needed for the forecasting purpose
- The dataset was checked for missing values, infinite values, and inconsistencies. Any invalid observations were removed to ensure data quality and reliability.
- The cleaned dataset containing the weekly RON95, RON97, and diesel fuel prices was kept for the next steps of time series analysis and forecasting.

4.2 TIME SERIES ANALYSIS AND ARIMA MODELLING

4.2.1 Time Series Visualization

The cleaned fuel price data was plotted using line charts to observe how prices changed over time. Time series plots were generated to identify possible trends, fluctuations, and structural changes in the RON95 fuel price series.

4.2.2 Descriptive Analysis

Summary statistics including the mean, standard deviation, minimum value, and maximum value were calculated to provide an overview of the characteristics and variability of the RON95 fuel price series.

4.2.3 Data Partitioning

To test the forecasting performance, the dataset was split into training and testing sets. Around 80% of the data was used for training the model, while the remaining 20% was used to test and evaluate the forecast results.

4.2.4 Stationary Testing

Stationarity is a fundamental requirement for ARIMA modelling. The Augmented Dickey-Fuller (ADF) test was applied to determine whether the RON95 series is stationary. The null hypothesis of the ADF test states that the series contains a unit root and is therefore non-stationary. In addition, the series was visually inspected using plots to assess the presence of trends, seasonality, and other patterns that may indicate non-stationarity.

4.2.6 Model Identification and Estimation

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the original RON95 training series were examined to identify suitable autoregressive (p) and moving average (q) orders. Several models were tested and compared, including ARIMA(1,0,1), ARIMA(2,0,1), ARIMA(1,0,2), and ARIMA(2,0,2).

4.2.7 Model Selection

The candidate ARIMA models were compared using the Akaike Information Criterion (AIC). The model with the lowest AIC value was chosen as the best model because it offers a good balance between fit and simplicity.

4.2.8 Diagnostic Checking

Residual checks were done to see how well the chosen ARIMA model performed. Residual plots and residual ACF plots were examined to determine whether any significant autocorrelation remained. The Ljung-Box test was also conducted to formally test whether the residuals behaved as white noise.

4.2.9 Forecasting and Accuracy Evaluation

The selected ARIMA model was used to generate forecasts for the testing dataset. Forecast accuracy was measured using the Mean Absolute Error (MAE) , Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). The best model was then refitted using the full dataset to produce a 12-week-ahead forecast of future RON95 fuel prices.

5.0 RESULTS AND DISCUSSION

5.1 Descriptive Statistics

	Mean	Std	Min	Max
RON95	2.105	0.363317	1.25	4.27

Figure 1 : Descriptive Statistics

The average RON95 price was RM2.105 per litre with a standard deviation of 0.363. Prices ranged from a minimum of RM1.25 to a maximum of RM4.27, noticeable fluctuations over time.

5.2 Time Series Plot and Decomposition

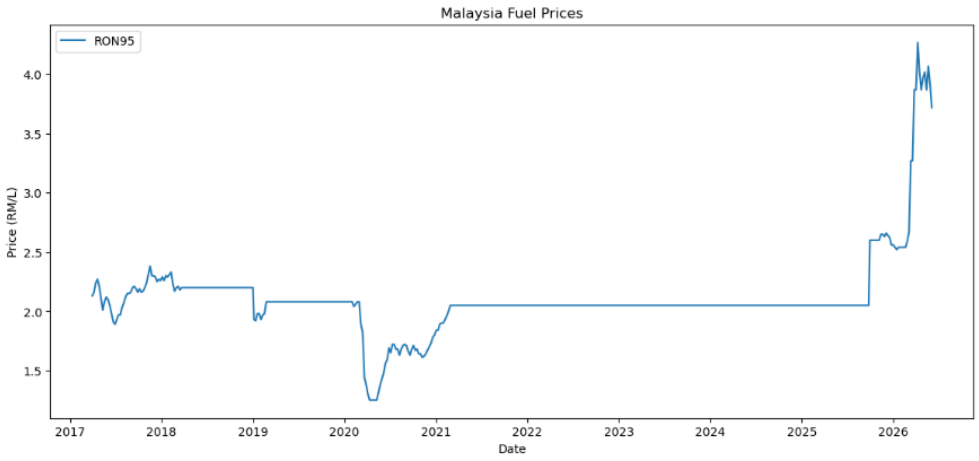


Figure 2 : Time Series Plot

The RON95 price series exhibits periods of fluctuation and stability. Prices fell sharply in 2020, remained relatively constant from 2021 to 2025, and increased significantly in 2026, reaching a peak of approximately RM4.27 per litre before declining slightly.

The time series does not appear to have a strong long-term upward or downward trend for most of the study period. However, the sharp increase observed in 2026 suggests the presence of structural changes and changing variability over time, which should be further examined using the Augmented Dickey-Fuller (ADF) test.

5.3 Stationarity Test Results

```
 RON95 Training
ADF Statistic: -3.1544512264459925
p-value: 0.02277730146230277
Stationary
```

Figure 3 : ADF Test

The ADF test produced a statistic of -3.154 and a p-value of 0.02278. As the p-value is below 0.05, the null hypothesis of a unit root is rejected, indicating that the original series is stationary and does not require differencing.

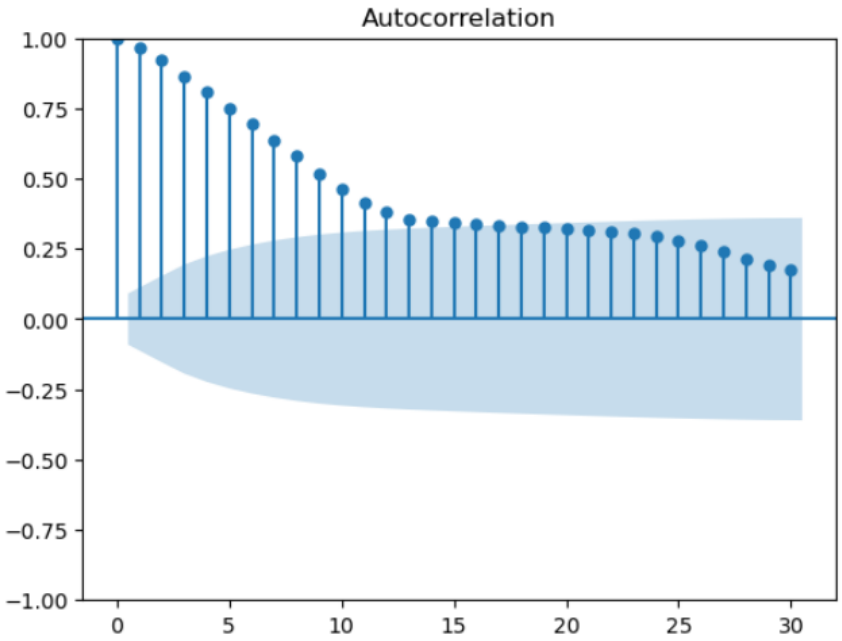


Figure 4 : ACF Plots

The ACF plot shows a gradual decline in autocorrelation values as the lag increases. Most of the autocorrelations remain outside the confidence bounds, indicating that the RON95 price series exhibits strong correlation with its past values. This slow decay pattern suggests the presence of an autoregressive structure in the data and indicates that previous observations have a significant influence on current prices.

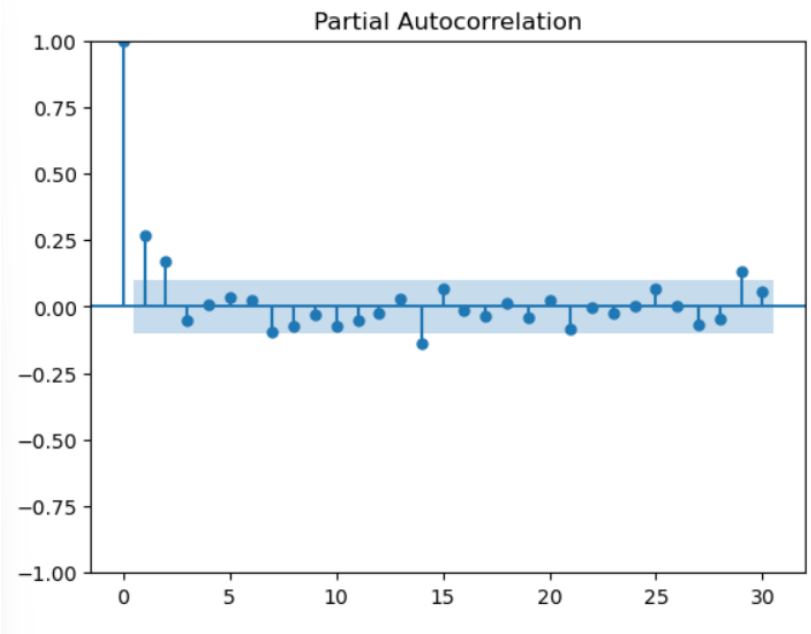


Figure 5 : PACF Plots

The PACF plot displays a significant spike at lag 1, while most of the subsequent lags fall within the confidence bounds. This indicates that the first lag has a strong direct effect on the current value of the series after accounting for the influence of intervening lags. The pattern suggests that a first-order autoregressive component, AR(1), may be appropriate for modelling the RON95 price series.

5.4 Arima Model Identification and Estimation

```
ARIMA(1,0,1): -1433.2155667912098
ARIMA(2,0,1): -1451.9191444454648
ARIMA(1,0,2): -1453.2919763421653
ARIMA(2,0,2): -1452.3998102114479
```

Figure 6 : Compare AIC

ARIMA(1,0,2) was selected because it produced the lowest AIC value among the candidate models, indicating that it provides the best balance between model accuracy and parsimony.

SARIMAX Results

Dep. Variable:	ron95	No. Observations:	371
Model:	ARIMA(1, 0, 2)	Log Likelihood	731.646
Date:	Fri, 19 Jun 2026	AIC	-1453.292
Time:	20:46:36	BIC	-1433.711
Sample:	0	HQIC	-1445.515
	- 371		
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
const	2.0287	0.094	21.471	0.000	1.843	2.214
ar.L1	0.9631	0.012	78.465	0.000	0.939	0.987
ma.L1	0.2351	0.044	5.314	0.000	0.148	0.322
ma.L2	0.2460	0.024	10.406	0.000	0.200	0.292
sigma2	0.0011	2.86e-05	39.285	0.000	0.001	0.001

Ljung-Box (L1) (Q):	0.10	Jarque-Bera (JB):	24756.37
Prob(Q):	0.75	Prob(JB):	0.00
Heteroskedasticity (H):	0.00	Skew:	-4.43
Prob(H) (two-sided):	0.00	Kurtosis:	42.03

Figure 7 : Model Summary

5.5 Residual Diagnostics

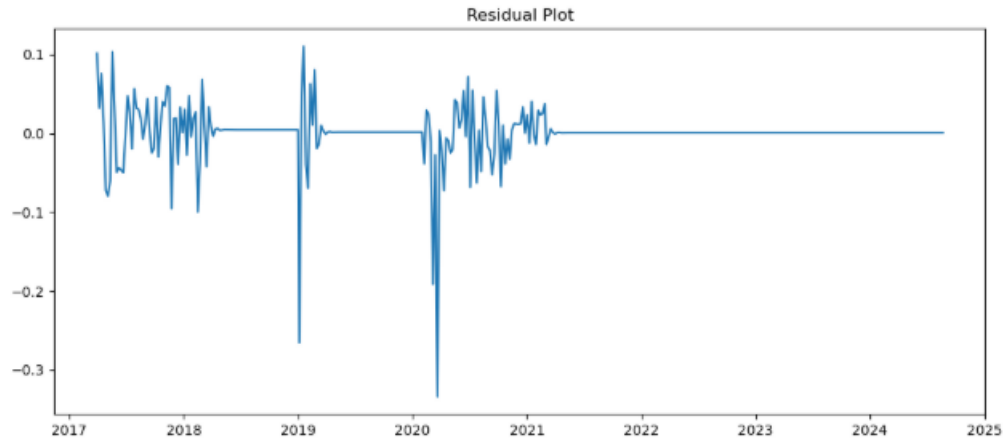


Figure 8 : Diagnostic Plot Residual Over Time

The residual plot shows that the residuals fluctuate randomly around zero with no clear trend or systematic pattern. This suggests that the ARIMA model has successfully captured the underlying structure of the RON95 price series. Although a few large residuals are observed, particularly around 2019 and 2020, the residuals are generally centered around zero, indicating an adequate model fit.

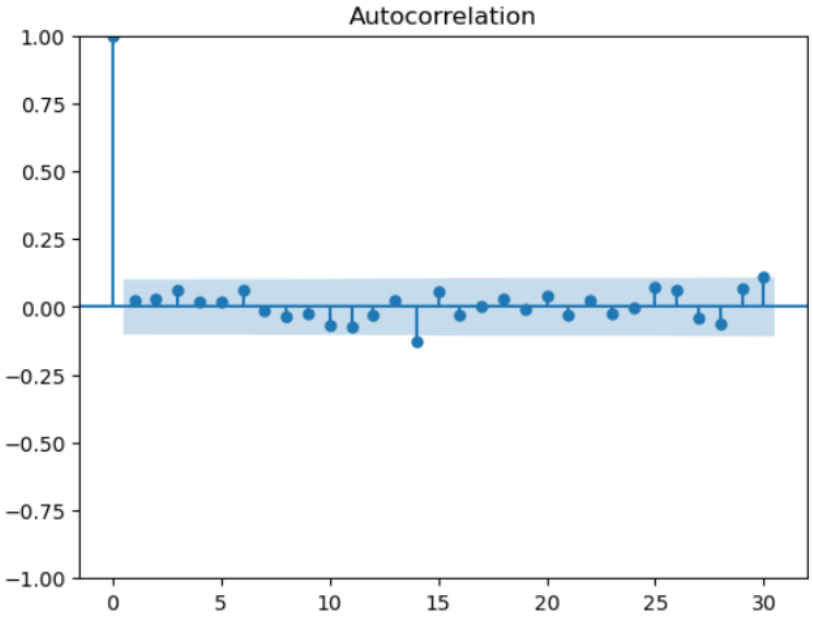


Figure 9 : Diagnostic Plots Acf of Residuals

Most residual autocorrelations fall within the confidence bounds, indicating that the residuals resemble white noise and that the model has adequately explained the serial dependence in the data.

```
(371,)
     lb_stat  lb_pvalue
10  6.372522   0.783055
```

Figure 10 : Ljung-Box Test

As p value is > 0.05 , The Ljung-Box test was performed to examine whether the residuals were independently distributed. The test produced a p-value of 0.783055, which is greater than the significance level of 0.05. Therefore, the null hypothesis cannot be rejected, indicating that the residuals behave as white noise. This suggests that the ARIMA model adequately captures the underlying pattern of the fuel price series and is suitable for forecasting.

5.6 Forecast Results and Accuracy

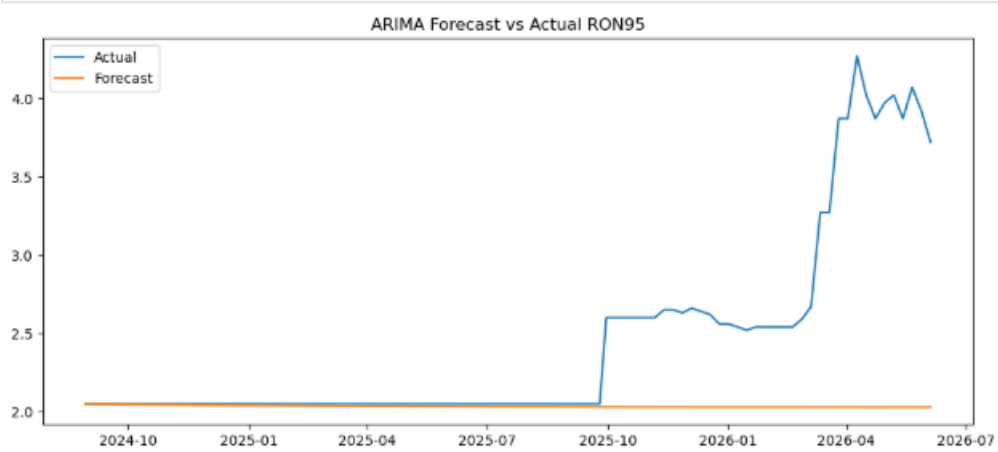


Figure 11 : Forecast Plot Comparing The Model's Forecast for The last 12 weeks

```
MAE  : 0.4004
RMSE: 0.7421
MAPE: 12.2818 %
```

Figure 12 : Forecast Accuracy Measures on The Test Set (Last 12 weeks)

The forecasting performance of the selected ARIMA model was evaluated using MAE, RMSE and MAPE. The model achieved an MAE of 0.4004, RMSE of 0.7421 and MAPE of 12.28%. These results indicate that the model provides reasonably accurate forecasts for RON95 prices, with an average percentage forecasting error of approximately 12.28%. According to common forecasting accuracy guidelines, a MAPE value between 10% and 20% indicates good forecasting performance.

5.7 Future Forecast (12 weeks ahead)

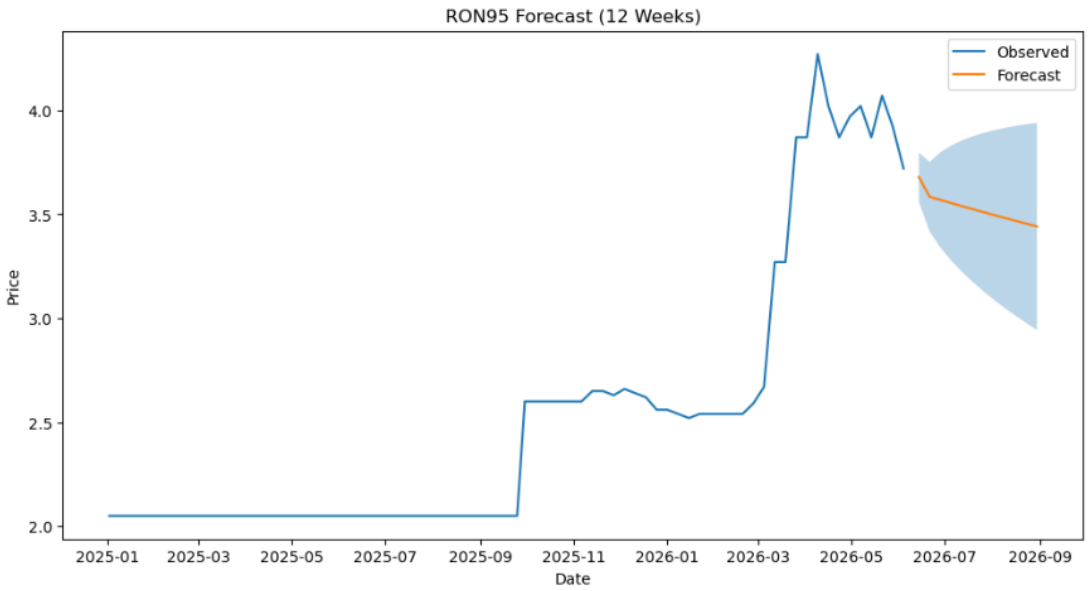


Figure 13 : 12 Weeks Ahead Forecast Plot

The best ARIMA model was refitted using the complete dataset and used to forecast future RON95 prices. The forecast suggests that RON95 prices are expected to exhibit a slight downward trend over the next 12 weeks, decreasing gradually throughout the forecast period. The widening confidence intervals indicate increasing uncertainty as the forecast horizon extends.

6.0 CONCLUSION

This study analysed the historical trends of weekly RON95 fuel prices in Malaysia using time series analysis and ARIMA modelling. The ADF test confirmed that the series was stationary, while ACF, PACF and AIC comparisons identified ARIMA(1,0,2) as the most suitable forecasting model. Residual diagnostics and the Ljung-Box test indicated that the model was adequate and that the residuals behaved as white noise. The model achieved an MAE of 0.4004, RMSE of 0.7421 and MAPE of 12.28%, demonstrating good forecasting performance. The 12-week forecast suggested a slight downward trend in RON95 prices. Overall, the selected ARIMA model proved effective for forecasting Malaysia's weekly RON95 fuel prices.

7.0 References

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